Project 5: Image Identification with Teachable Machine Tool

Overview

The training data for the AI model is very crucial. The project lets students feed the training data to the AI model and test it with a fresh data set. The digital tool will demonstrate the working principle of the AI models with a simpler interface.

Learning Outcomes

- *AI Model Training.* The simple interface tool will let the students experience the training of AI model
- *Input Training Data and Testing.* The input images of two categories will be feeded to the AI model. The quality of input images (similarity and non-similiarity of the data) will be reflected during testing the AI model.

Material Required

- Computer (preferable connected with projector) or smart board
- Printed input and test images

Project Plan

Project Activity	Time	Description
Launch	10-15 mins	The teacher will demonstrate the Teachable Machine tool to students. The simple tool will help students visualize the training process for AI models.
Experience	20-25 mins	After training, the students will show the test images to let trained AI model identifies the input images.
Explore	1 week	Students work with various training data-set of their own and compare their results.

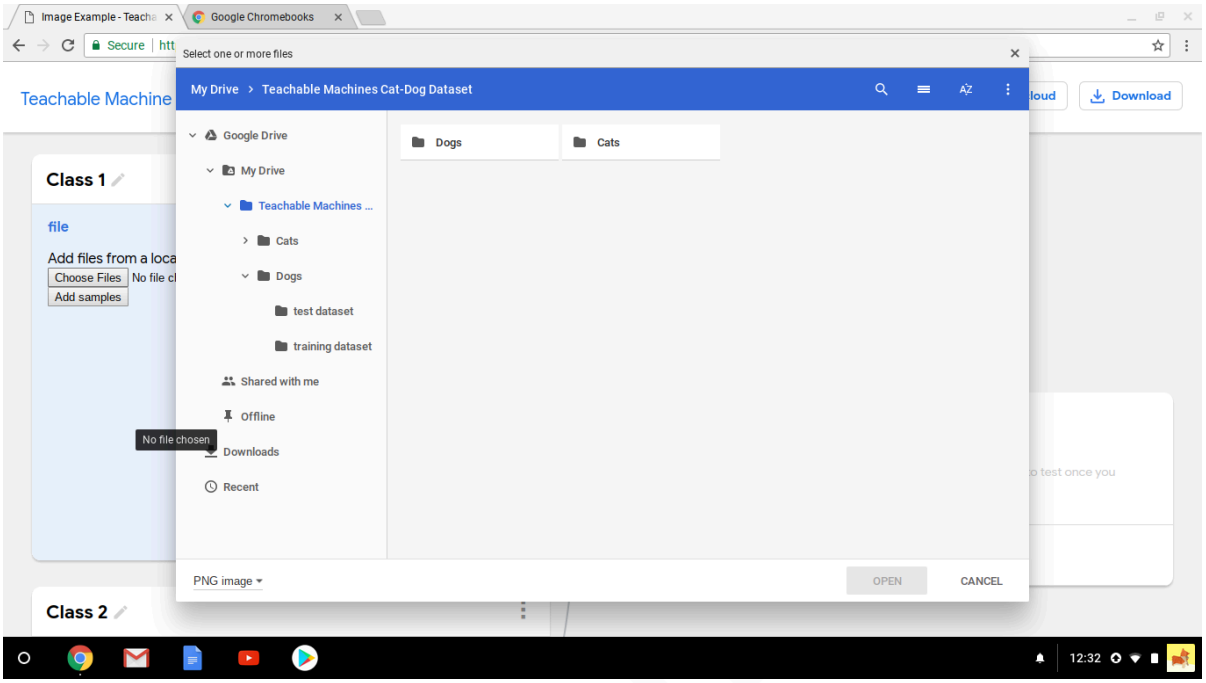
Project

- Go to the website: https://teachablemachine.withgoogle.com. Click *"Get Started,"* then click *"Image Project", and then select the "Standard Image Model"* option.
- A set of images is given at the end to build the cat-dog classifiers. Group the examples of images of dogs and cats into classes that we want the computer to learn.
- Two sets of datasets will be used:
    - *Initial Training Dataset* - Images to "teach" the AI model model which image is a cat and which image is a dog. A diverse set of images will help the AI to identify the images more accurately.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

- *Test Dataset* - Images to test the AI model after training. These images will not be used while teaching AI model.

- You may take the prints of these images and show them to the camera for training.
- Compare the test results for various input images.



Explore

1. For the dog training dataset, record the following:
    - How many images are included?
    - How are the images similar?
    - How are the images different?

2. For the cat training dataset, record the following:
    - How many images are included?
    - How are the images similar?
    - How are the images different?

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

3. Once your classifier is finished, test your dataset with the images below. Fill in the table on the next page about your testing dataset:

Image	Classification	Confidence Score	Correct?

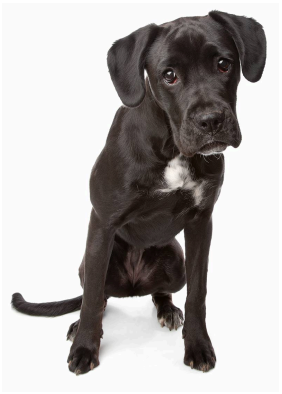
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Reference: AI + Ethics Curriculum for Middle School, Massachusetts Institute of Technology.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Training Dataset

(Dogs)



© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Training Dataset

(Cats)

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Test Dataset

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.